Optimal Dynamic Pricing With Preferences for Novelty

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Motivation

- Firms have more access than ever to consumer information
- Allows for targeted discounting based on preferences derived from repeat interactions
- Such discounting feasibly allows firms to steer consumers toward certain products in pursuit of profit maximization

How do firms price based on repeated interactions with consumers and how are profits affected?

Research Question

How can firms price discriminate via consumer preferences for novelty and habit and how much is information on said preferences worth?

How do I answer?

- Solve a theoretical model to understand firm's behavior
- Estimate consumer's preferences with Nielsen data, recovering the distribution of novelty/habit preferences
- Fit that distribution to the supply theory to understand how firm's revenues are impacted by this sort of targeting

Literature

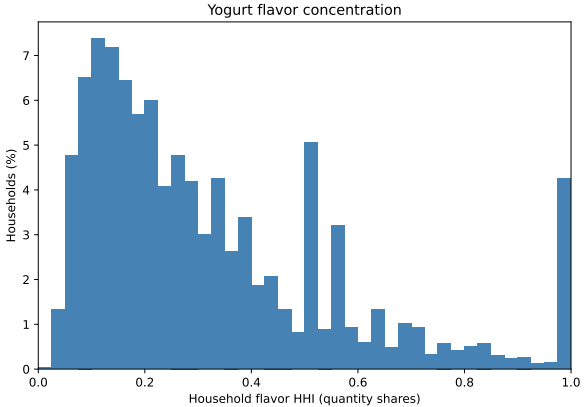
Targeting:

- Dubé (2008)
- Dubé et. al (2022)
- Anand & Shachar (2009)
- Karle & Reisinger (2024)
- Durrmeyer, Fournel, Samano (2026)

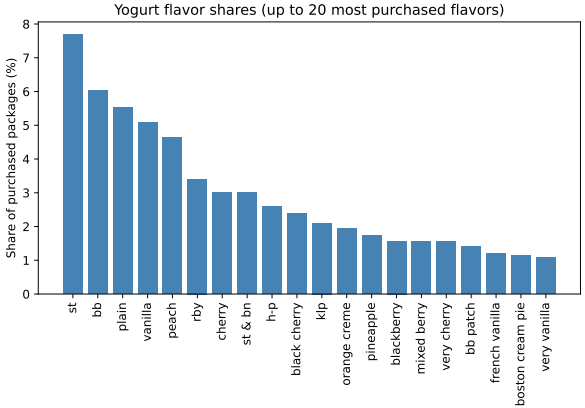
Estimating preferences for novelty:

- Dubé (2004)
- Dubé et. al (2021)
- Chintagunta (1999)
- Lattin (1987)
- McAllister (1982)
- Ackerberg (2003)

HHI


Yogurt flavor concentration
Households (%)
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
1.0
Household flavor HHI (quantity shares)

Flavor Popularity



Supply Setting

- Monopolistic firm producing a single product j
- Firm is profit maximizing, setting prices according to that objective
- A consumer is born into the world with a type $r \in \{H, L\}$, determining how much utility they derive from switching and an initial state $h_0 \in \{0, j\}$
- Consumers live for two periods, purchasing the inside good or opting for the outside good in each period
- The firm knows initial states and types, setting prices to individual consumers by type and history

Consumer's Problem

$$V_{jt}^{inside} = \underbrace{\beta}_{\text{Constant}} + \overbrace{\gamma_r \mathbb{1}\{h=0\}}^{\text{Utilty from Switching}} - \underbrace{\alpha p_t(h)}_{\text{Price Disutility}}, \quad V_{0t}^{Outside} = \underbrace{\gamma_r \mathbb{1}\{h=j\}}_{\text{Utility from Switching}}$$

$$v_t(x) = V_{jt}^{inside} - V_{0t}^{outside} = \beta + \underbrace{\gamma[\mathbb{1}\{h=0\} - \mathbb{1}\{h=j\}]}_{q(h)} - \alpha p_t(h)$$

- Prices, set by the firm, and switching utility depend on history $h = x_{t-1}$
- Utility for the outside good is normalized to zero

Firm's Problem

$$\pi_t(h) = \max_{p_t(h)} s_t(h)[p_t(h) - c]$$

$$\Pi(h) = \underbrace{s_1(h)(p_1(h) - c)}_{\text{Margin in Period 1}} + \underbrace{\delta\{s_1(h)\pi_2(j) + (1 - s_1(h))\pi_2(0)\}}_{\text{Continuation Value}}$$

- Firm profits are comprised of period one margins, given history of consumption, and the value of that consumer in the different states in period two
- The firm will set prices according to consumer type and history to maximize expected profit

Optimal Terminal Price and Profit

$$\frac{\partial \pi_2(h_1)}{\partial p_2(h_1)} = s_2(h_1) + (p_2 - c)\frac{\partial s_2(h_1)}{\partial p_2(h_1)}$$

$$p_2^*(h_1) = c + \frac{1}{\alpha(1 - s_2^*(h_1))}$$

$$\pi_2^*(p_2^*(h_1); h_1) = \frac{s_2^*(h_1)}{\alpha(1 - s_2^*(h_1))}$$

- Optimal period two price is cost plus a demand markup term
- Period two profit at optimal prices is the odds of purchase, scaled by the inverse price sensitivity

First Period Optimization

$$\frac{d\Pi}{dp_1(h_0)} = s_1 + \frac{\partial}{\partial p_1(h)} s_1(h_0)[p_1(h_0) - c + \delta(\pi_2(j) - \pi_2(0))]$$

$$p_1^*(h_0) = c + \frac{1}{\alpha(1 - s_1^*(h_0))} - \delta(\pi_2(j) - \pi_2(0))$$

$$\Pi^*(h) = \delta\pi_2(0) + \frac{s_1^*(h_0)}{\alpha(1 - s_1^*(h_0))}$$

Optimal Profit Given Switching Type

$$\frac{d\Pi}{d\gamma} = \frac{q(h)s_1^*(h) - \delta s_1^*(h)s_2^*(j) + \delta(1 - s_1^*(h))s_2^*(0)}{\alpha}$$

- There are two interpretations:
  1. If $h_0 = j$, $q(h) < 0$, leading us to ambiguous signage as we raise profits from initially outside consumers but see opposing forces for the initially inside: leave in period one, return in period two
  2. If $h_0 = 0$, $q(h) > 0 \Rightarrow \frac{d\Pi}{d\gamma} > 0$, meaning that profits increase as γ increases
- Points to firms potentially reaping great benefit from knowing consumer types and histories

Pricing Simulations Specification

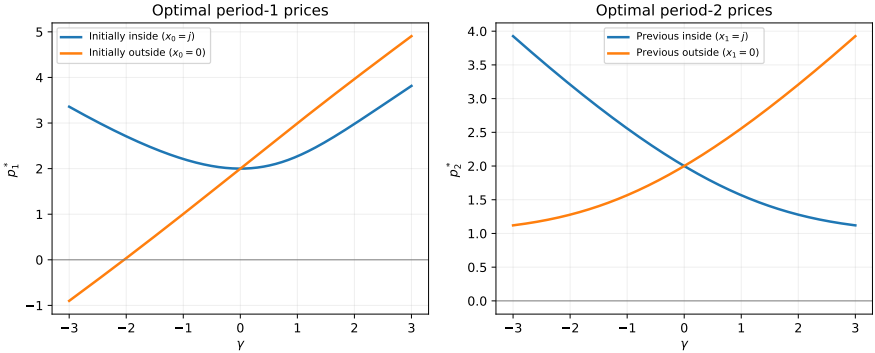
Parameter	Description	Value
β	Inside-good intercept	2
α	Price sensitivity	1
δ	Discount factor (no discounting)	1
c	Marginal cost	0
γ	Switching preference	$[-3, 3]$
Initial Consumption State		$h_0 \in \{0, j\}$
Feasible Prices		$p_t(h) \in \mathbb{R}$

- Allow for negative pricing
- Firm has knowledge of both type and history, allowing for targeted pricing
- Four consumption histories possible $(h_0, h_1) \in \{(0,0), (0,j), (j,j), (j,0)\}$

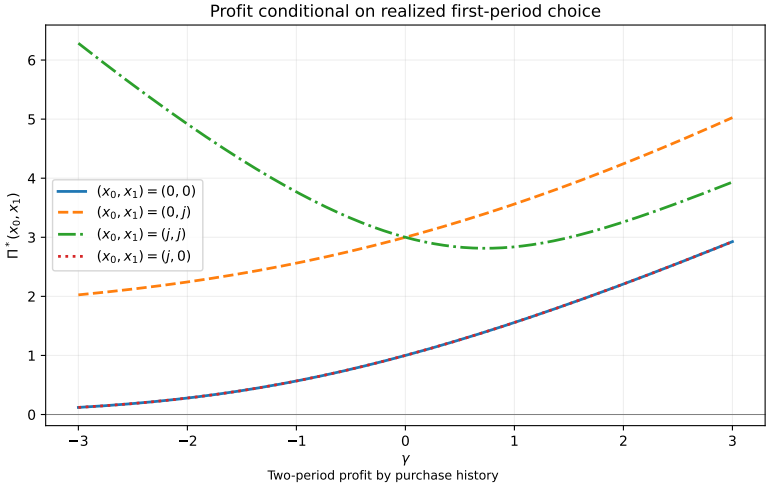
Results

x_0	γ	p_1^*	$p_2^*(j)$	$p_2^*(0)$	Π^*	$\Pi^{*\prime}$
j	-3	3.358	3.926	1.120	5.284	-1.445
j	-1	2.211	2.557	1.567	2.768	-0.993
j	0	2.000	2.000	2.000	2.000	-0.500
j	1	2.271	1.567	2.557	1.838	0.177
j	3	3.814	1.120	3.926	2.934	0.730
0	-3	-0.901	3.926	1.120	1.026	0.177
0	-1	1.005	2.557	1.567	1.562	0.376
0	0	2.000	2.000	2.000	2.000	0.500
0	1	2.995	1.567	2.557	2.562	0.624
0	3	4.905	1.120	3.926	4.025	0.823

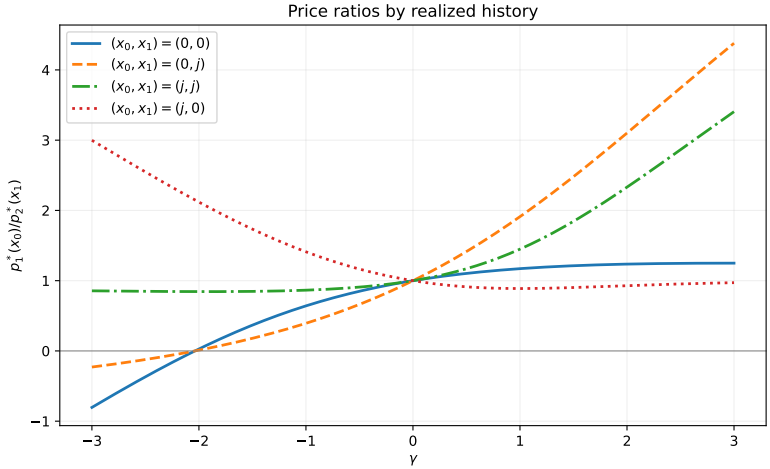
Optimal Prices



Optimal Profit



Optimal Price Ratio



Discussion

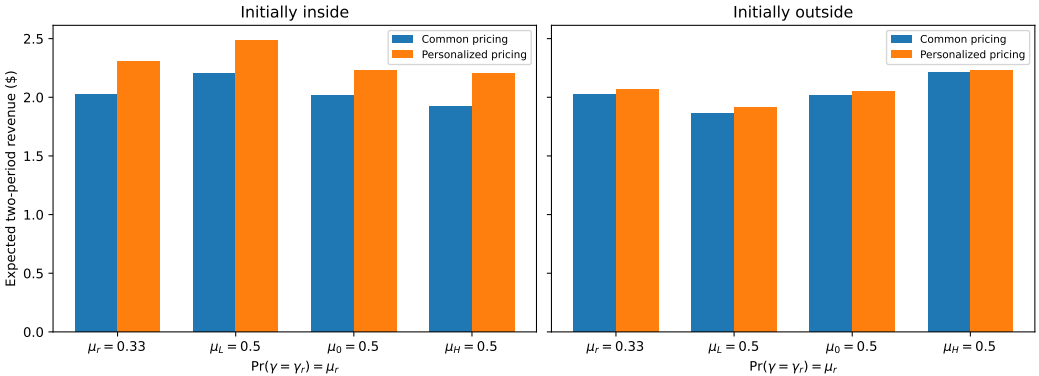
- Two mechanisms: Induce and harvest, Retain and gain
- Most profit from those who are habitual inside purchasers
- Least profit from those who purchase outside in period 1 across all levels of gamma
- Interesting when (j, j) and $(0, 0)$ cross

Information Friction: Fixed Beliefs about Switching Type

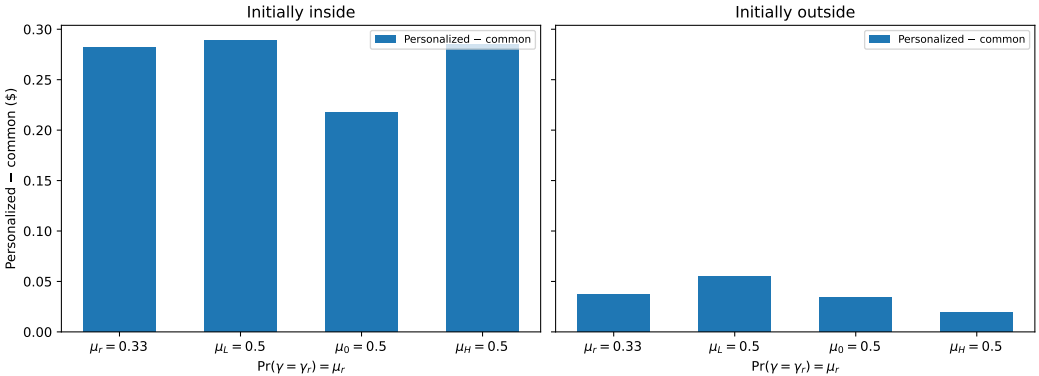
- Initial purchase history is observed.
- $\gamma_L = -1.25$, $\gamma_0 = 0$, $\gamma_H = 1.25$.
- $\Pr(\gamma = \gamma_r) = \mu_r$, $r \in \{H, 0, L\}$. Equal case: $\mu_r = 0.33$.
- Other cases: each permutation of $(0.50, 0.25, 0.25)$.
- Type probabilities stay fixed in both periods, after either purchase outcome.
- Offers depend on observed history; there is no learning about type.

$$A(p, h; \lambda) = \sum_r \lambda_r s_r(p, h), \qquad W(a; \lambda) = \max_x A(x, a; \lambda)(x - c).$$

Known versus Uncertain Switching Type



Value of Observing Switching Type



Data Sources

Nielsen HomeScan

- Provides household level purchase data
- Covers years 2006-2015
- Products chosen, quantity chosen, timing of trips

Nielsen RetailScan

- Provides store level product data
- For specified product category: prices, markets, product attributes

Sample Description

- Must be single-person household
- Must be seen at least 2 times
- Purchases must be made at Nielsen store (prices available)
- Flavor attribute must be present

Sample Statistics: Purchase Activity

	Full sample	Yogurt sample
Households	37,827	15,106
Observed panel years	3.96	3.16
Trips per household	493.13	20.26
Trips per observed week (approx.)	2.27	0.11
Distinct UPCs per household	1,178.31	13.89
Purchase lines per household	3,397.64	45.15
Packages per household	4,669.07	93.77

All statistics except household counts are means.

Sample Statistics: Demographics

	Full sample	Yogurt sample
Head age (category code)	6.96	7.01
Household income (category code)	16.18	17.06
White (%)	80.33	81.58
Black (%)	12.81	12.15
Asian (%)	1.57	1.57
Other race (%)	2.19	2.11
Hispanic (%)	3.10	2.59

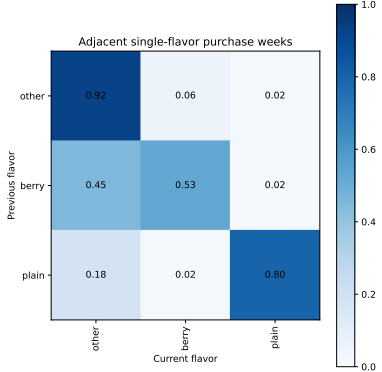
Age and income are mean category codes.

Switching Stats

Table: Matched yogurt sample: purchase dynamics

metric	value
Matched households	15106.00
Observed yogurt purchase weeks	282648.00
Mean flavors bought per household	10.76
Mean household flavor HHI (0 to 1)	0.32
Mixed-flavor weeks (%)	37.88
Weeks above seven packages (%)	21.72
Eligible adjacent single-flavor week pairs	30392.00
Switch rate in eligible pairs (%)	46.75
Mean single-flavor spell length (consecutive calendar weeks)	1.10
Median gap between purchase weeks	3.00

Persistence Across Aggregated Flavor Groups



Plain repeats in approximately 80% of eligible transitions, versus 53% for berry.

Demand Setting and Objective

- Consumers can either purchase the inside good, a flavor of yogurt, or the outside good, not yogurt
- Following Nevo (2001), a week contains seven choice opportunities with observed flavor purchases treated as fulfilling an opportunity
- Define outside good as $\max(0, 7 - \sum_j x_{jt})$, or the difference in purchases and a full week
- Consumers differ in preferences for novelty and price sensitivity, but derive the same base utility from the three flavor categories
- Goal: Recover distribution of novelty preferences

Demand Model

$$U_{ijt} = \overbrace{\beta_0}^{Constant} + \underbrace{\beta_j X_{jt}}_{\text{Flavor specific constant}} + \overbrace{\gamma_i N_{ijt} + \phi_i H_{ijt}}^{\text{Preferences for novelty/habit}} + \underbrace{\alpha_i p_{jt}}_{\text{Price disutility}} + \overbrace{\varepsilon_{ijt}}^{\text{T1EV}}$$

$$\text{s.t.} \quad w_{ijt} = y_{ijt} / \sum_k y_{ikt}, \qquad H_{ijt} = \sum_{s<t} w_{ijs},$$

$$N_{ijt} = \begin{cases} 1 - w_{ij,t-1}, & \text{prior calendar week observed,} \\ 0, & \text{otherwise.} \end{cases}$$

- N_{ijt} is the prior week's purchase share in flavors other than j
- H_{ijt} accumulates prior weekly own-flavor purchase shares
- Normalize outside option to zero s.t. $U_{i0t} = \varepsilon_{i0t}$

Probability and Shares

$$P_{ijt} = \frac{\exp\{V_{ijt}\}}{1 + \sum_{k \in J} \exp\{V_{ikt}\}}$$

$$\text{s.t. } V_{ijt} = U_{ijt} - \varepsilon_{ijt}$$

$$s_{jt} = \frac{1}{N} \sum_{n=1}^{N} P_{ijt}$$

- Use deterministic utility to form purchase probabilities
- Take average probability for product j across individuals $i \in N$ to recover shares

Estimation

$$P_i(\theta) = \int P(y_i \mid \nu, \theta)\ \phi(\nu)\, d\nu$$

$$\widehat{P}_i(\theta) = \frac{1}{128} \sum_{r=1}^{128} \prod_{t=1}^{T_i} P_(y_i \mid \nu_{ir}, \theta) \qquad \nu_{ir} \sim \mathcal{N}(0, I)$$

$$Q_{128}(\theta) = -\sum_{i=1}^{N} \log \widehat{P}_i(\theta)$$

- ν_i captures unobserved heterogreneity, y_{it} represents observed choice, $\theta = \{\beta, \gamma, \alpha\}$ is the vector of heterogeneous demand parameters
- 128 draws per household from a normal distribution
- 3 starts at different starting values, keep lowest objective
- Minimize negative log-likelihood

Estimates

Table: Demand estimates without price controls

Parameter	Estimate	Standard error
Inside constant β_0	1.2173	0.0035
Berry vs. other	-0.6241	0.0034
Plain vs. other	-0.2719	0.0054
Mean price coefficient $E[\alpha_i]$	-0.8940	0.0044
Mean novelty	-0.7732	0.0095
Mean habit	0.0369	0.0002
SD price coefficient $\mathrm{SD}(\alpha_i)$	0.9019	0.0099
SD novelty	0.8003	0.0085
SD habit	0.0783	0.0004

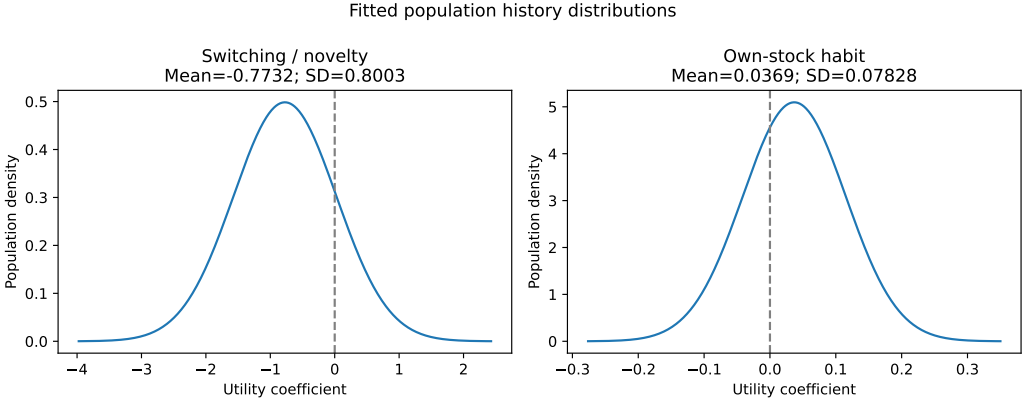
Implied Willingness to Pay

Table: Implied willingness to pay without price controls

Utility term	Mean WTP	Population SD
Inside constant β_0	2.7474	2.7716
Berry vs. other	-1.4085	1.4210
Plain vs. other	-0.6137	0.6192
novelty	-1.7451	3.1116
own stock	0.0833	0.2647

- Would need to be subsidized to switch, though lots of variation
- Willing to pay to continue consuming their same product

Distributions



Discussion

- Mean novelty effect is equivalent to $\approx \$1.75$ increase
- Habit provides some utility
- Both plain and berry are less preferred than other flavors

How can the firm use the distribution of switching types to price discriminate?

Connecting Supply and Demand

Supply Theory

- Today's pricing offer changes tomorrow's demand
- Single product firm pricing to attract customers against outside option
- Maintain two-period timing with initial state

Empirical Demand

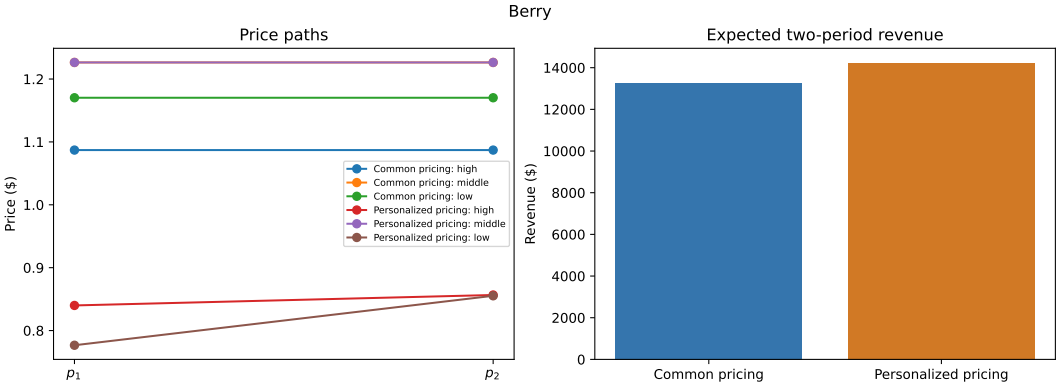
- Employ estimated preference for novelty
- Access observed purchase histories
- Novelty and habit both enter demand

How well does the theory map to what is observed in the data?

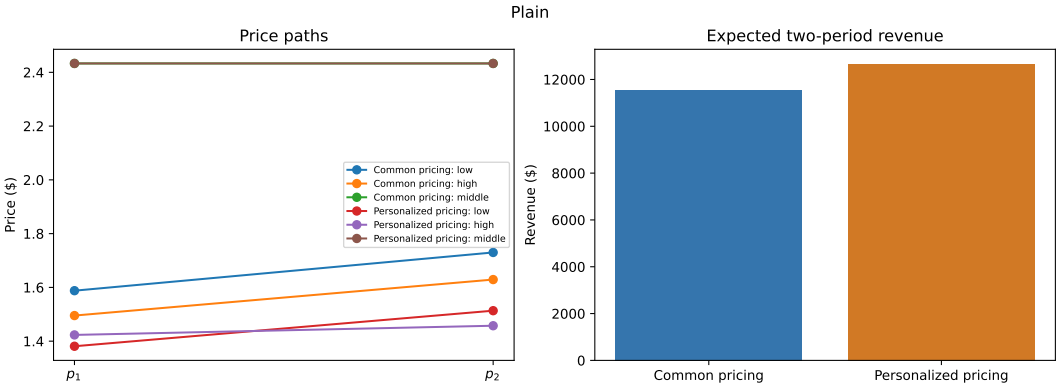
Procedure

- Recover household type from observed purchase history and treat estimated type as known to firm
- Use initial purchase y_{i1} to determine "initial state", h_0
- Separate high and low types as the 75th and 25th percentiles, respectively
- Use observed prices and simulate discounts based on them, specific to the different types and their consumption path
- Focus on berry and plain to maintain parity with the theoretical model and avoid aggregate weighting
- Use the controlled split fit and estimated type γ_i to discriminate, choosing the initial offer p_1 and then the terminal offer p_2 after observing the first-period choice

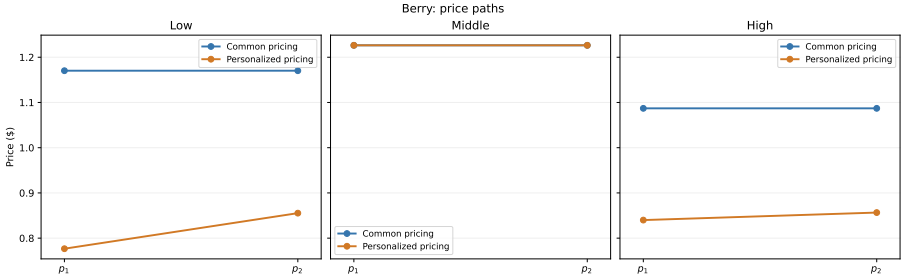
Results: Berry Inside



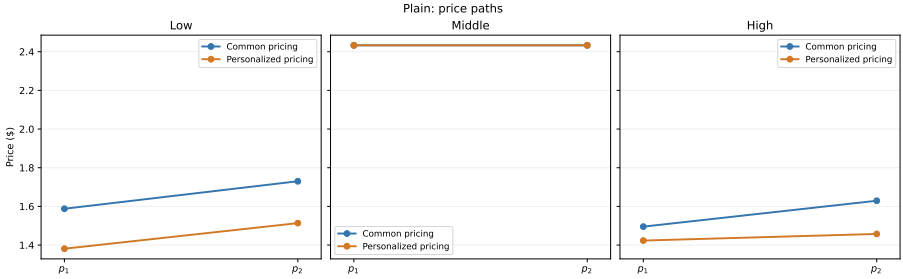
Results: Plain Inside



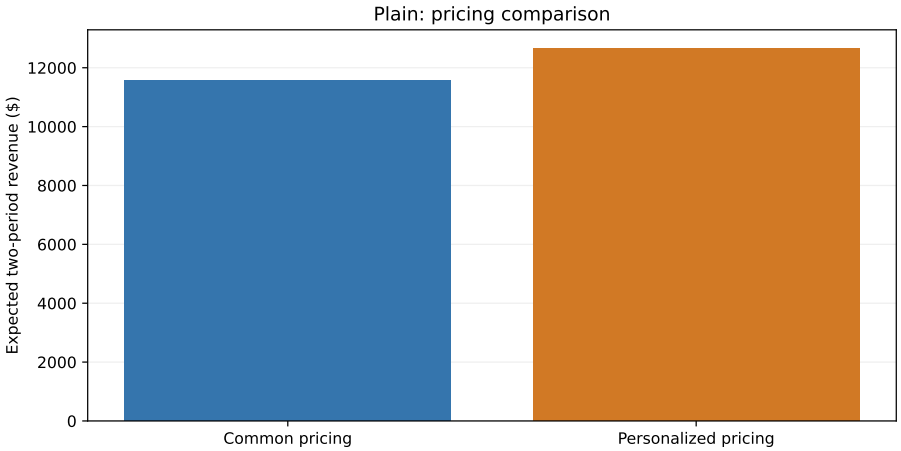
Berry: Optimal Offers by Consumer Type



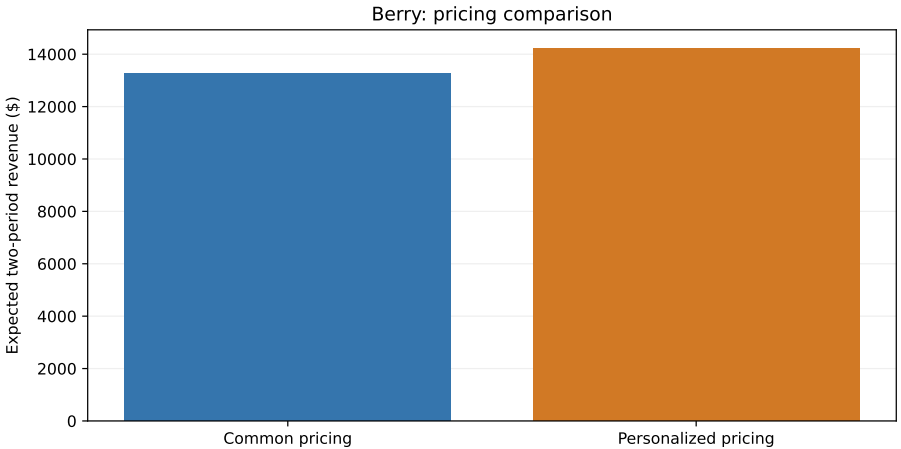
Plain: Optimal Offers by Consumer Type



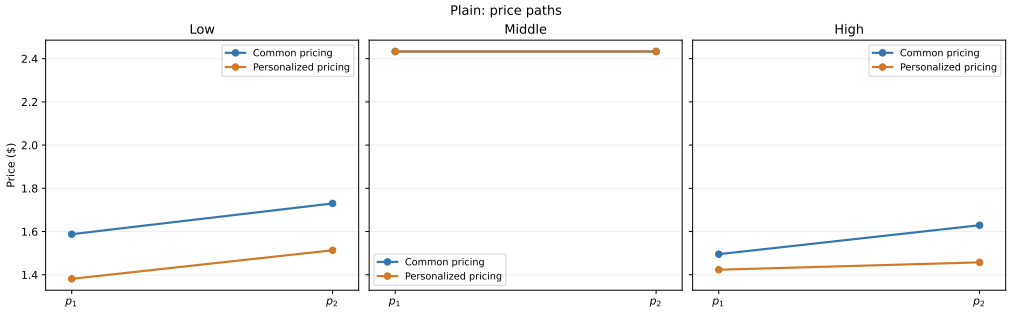
Pricing Comparison: Plain



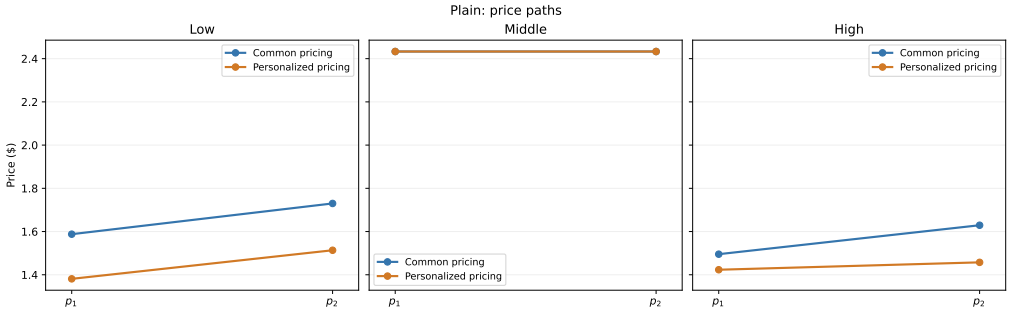
Pricing Comparison: Berry



Plain: Price Paths



Plain: Price Paths



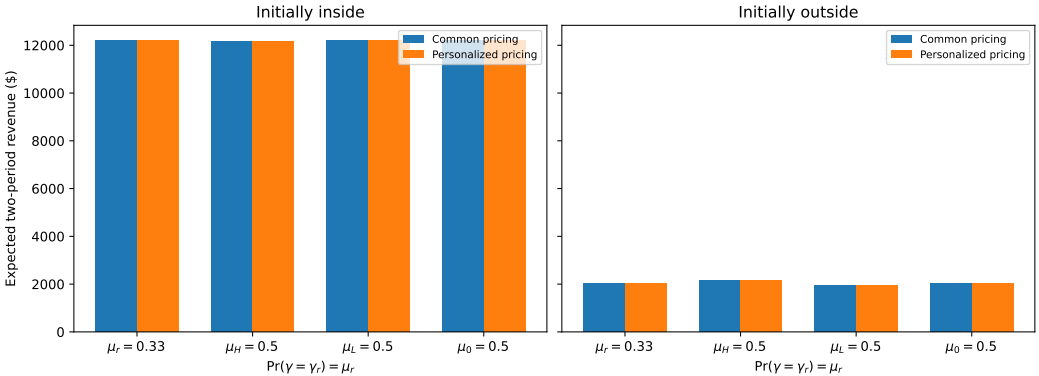
Comparison

Flavor	Policy	Revenue	Gain (%)
Plain	Common pricing	11,559.5	0.00
Plain	Personalized pricing	12,655.8	9.48
Berry	Common pricing	13,257.2	0.00
Berry	Personalized pricing	14,219.7	7.26

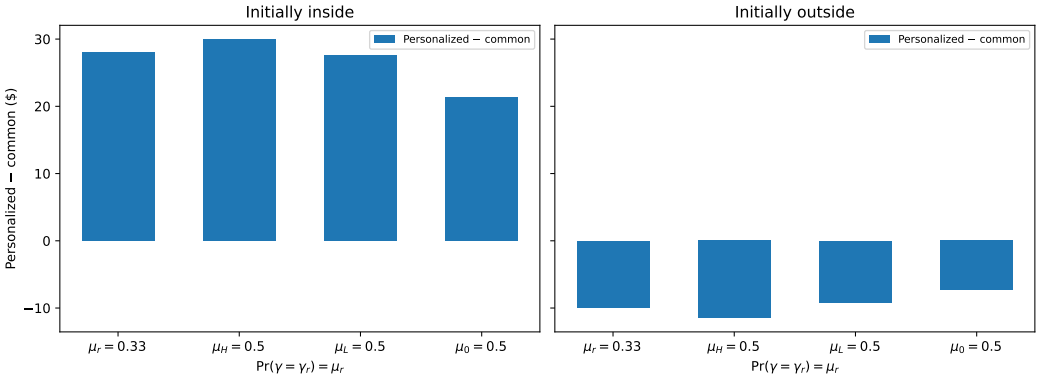
Empirical Type Information: Fixed Beliefs

- Low, middle, and high types use fitted group mean novelty coefficients.
- $\Pr(\gamma = \gamma_r) = \mu_r$. Equal case: $\mu_r = 0.33$; other cases permute $(0.50, 0.25, 0.25)$.
- Initial history is observed; type probabilities stay fixed after purchase.
- Evaluate both offer policies under the same persistent consumer types.
- Revenue gains use persistent-type revenue, not the fixed-belief objective.

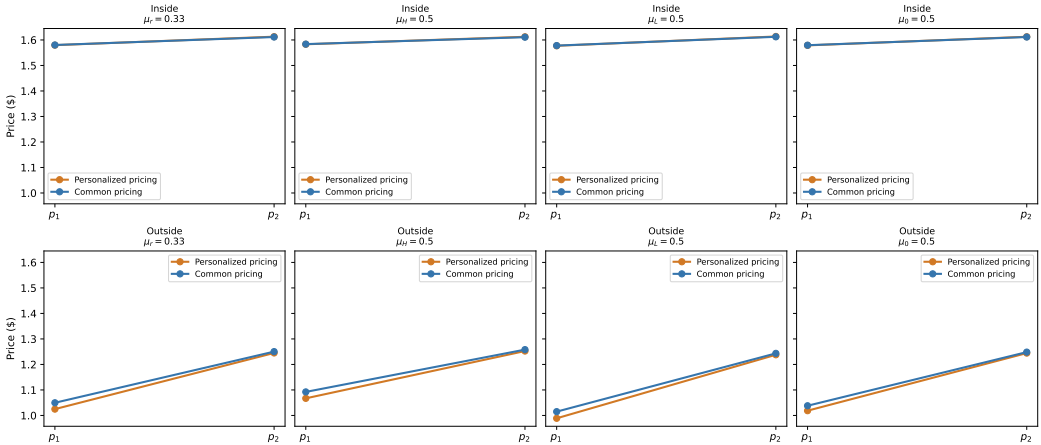
Plain: Revenue under Type Uncertainty



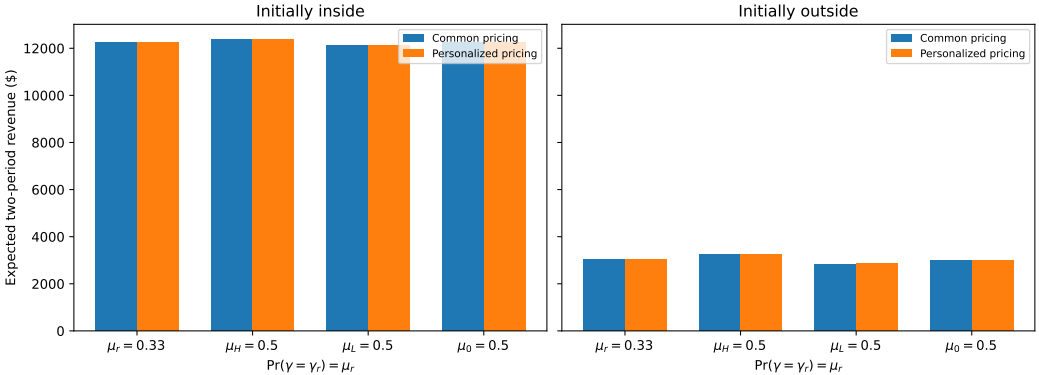
Plain: Revenue Value of Novelty-Type Information



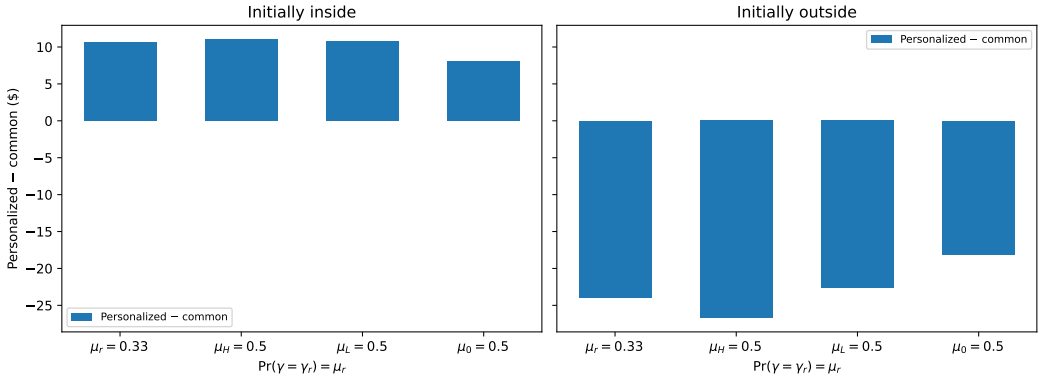
Plain: Price Paths under Type Uncertainty (p_1 to p_2)



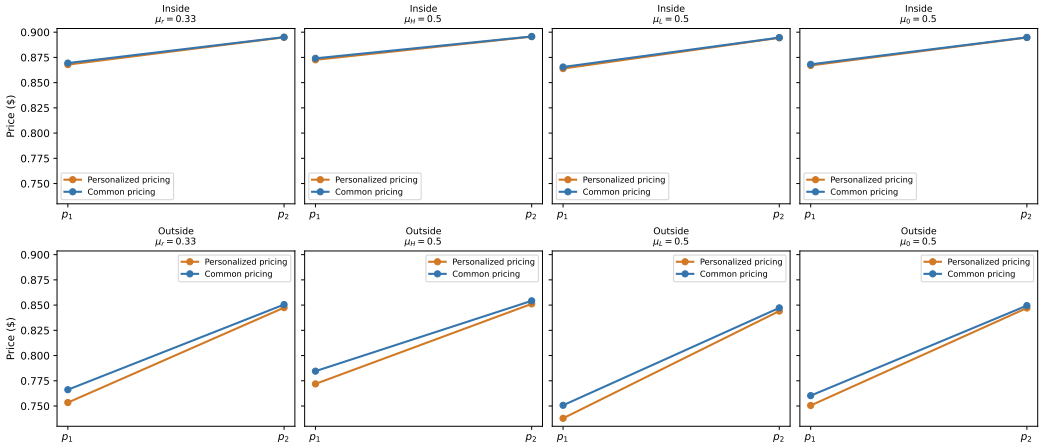
Berry: Revenue under Type Uncertainty



Berry: Revenue Value of Novelty-Type Information



Berry: Price Paths under Type Uncertainty (p_1 to p_2)



Conclusion

- Personalized pricing increases simulated revenue by up to 9.48% relative to common pricing
- Consumers experience disutility from switching, needing to be subsidized $\approx \$1.75$ to switch products
- Consumers enjoy habit and would be willing to pay $\$0.083$ to contribute another unit to the product's share of purchase history

Next Steps

- Add firm learning to supply side
- Extend market structure on supply
- Add more product attributes to empirical estimation
- Estimate unobserved quality in BLP fashion

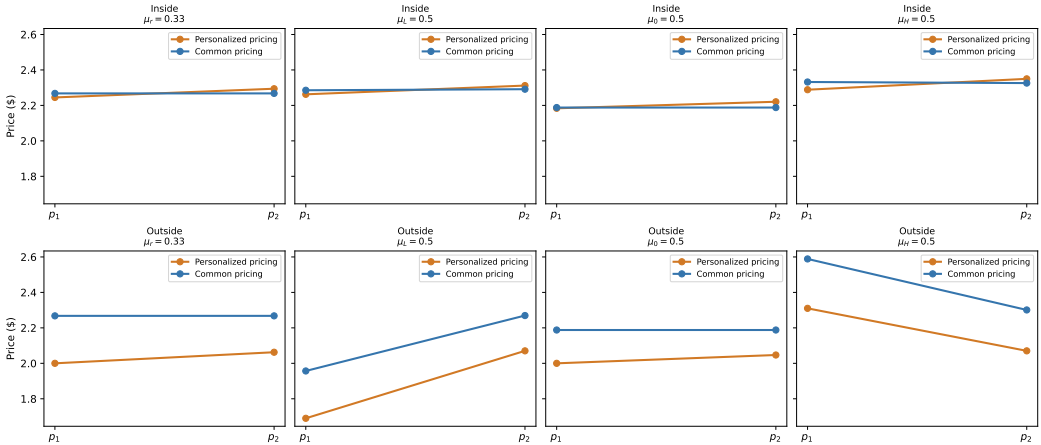
Shares and Elasticities

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \; s_{0t}(h) = 1 - s_t(h)$$

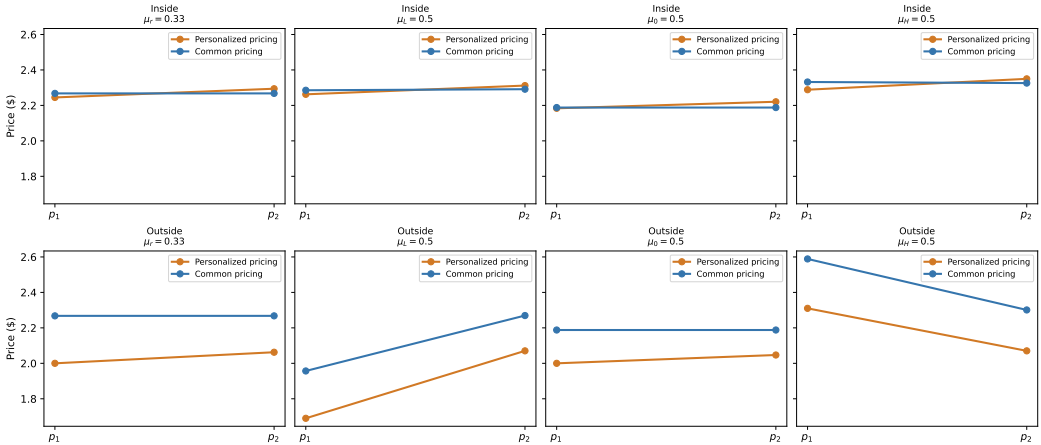
$$\frac{\partial s_t(h)}{\partial p_t(h)} = -\alpha s_t(h)(1 - s_t(h))$$

$$\frac{\partial s_t(h)}{\partial \gamma} = q(h) s_t(h)(1 - s_t(h))$$

Type Uncertainty: Price Paths



Type Uncertainty: Price Paths



Demand: Price-Control Comparison

Table: Demand estimates with and without price controls; standard errors in parentheses

Parameter	No controls	With controls
Mean novelty	-0.7732 (0.0095)	-0.7861 (0.0091)
Mean habit	0.0369 (0.0002)	0.0373 (0.0002)
SD novelty	0.8003 (0.0085)	0.7999 (0.0082)
SD habit	0.0783 (0.0004)	0.0793 (0.0004)

Controlled Demand Estimates

Table: Demand estimates with price controls

Parameter	Estimate	Standard error
Inside constant β_0	1.4762	0.0072
Berry vs. other	-0.6556	0.0034
Plain vs. other	-0.1192	0.0066
Mean price coefficient $E[\alpha_i]$	-1.0349	0.0051
Mean novelty	-0.7861	0.0091
Mean habit	0.0373	0.0002
SD price coefficient $\mathrm{SD}(\alpha_i)$	0.7957	0.0063
SD novelty	0.7999	0.0082
SD habit	0.0793	0.0004
Price control ρ	0.1959	0.0047

Controlled Fit: Price First Stages

Table: Price first stages for the controlled specification

Flavor	IV slope	Cluster SE	F	Partial R^2
Other	-34.680	0.966	1288.26	0.0213
Berry	-33.105	1.377	578.20	0.0192
Plain	-28.585	3.085	85.87	0.0076

Controlled Fit: Willingness to Pay

Table: Implied willingness to pay with price controls

Utility term	Mean WTP	Population SD
Inside constant β_0	2.2696	1.7451
Berry vs. other	-1.0081	0.7751
Plain vs. other	-0.1832	0.1409
novelty	-1.2086	1.8083
own stock	0.0574	0.1600